

Transported flags vanish on vectors generated by a trace vector

MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero

theorem

Turns decay of projection traces into norm convergence on each source-orbit vector.

Statement

Let $\sigma : A \rightarrow B(K)$ be a unital star representation of the completed CAR algebra on a complete complex Hilbert space. Suppose $\xi \in K$ implements the normalized trace from the CAR trace construction: $\langle \xi, \sigma(a)\xi \rangle = \tau(a)$ for all $a \in A$. For the transported projections $f_{j,n}$ of a fixed shell family and every $b \in A$, one has $\sigma(f_{j,n})\sigma(b)\xi \rightarrow 0$ in norm as $n \rightarrow \infty$.

Assumptions

Let A be the completed CAR algebra with matrix stages $M_{2^n}(\mathbb{C})$ embedded by $a \mapsto a \otimes I_2$. Write f_n for the image of the first diagonal matrix unit, $p_n = f_n - f_{n+1}$, and ω for the root state characterized by $\omega(\iota_n(a)) = a_{00}$ on each stage, where ι_n is its canonical embedding.

Index chosen pure-state Gelfand-Naimark-Segal (GNS) representations by their unitary-equivalence classes $j \in I$, with root index o and representative states φ_j .

A fixed shell family consists of complex-linear unital star automorphisms α_j and elements $w_{j,n} \in A$ satisfying $\varphi_j(\alpha_j(a)) = \omega(a)$, $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$ and $w_{j,n} w_{j,n}^* = p_n$.

At the root, α_o is the identity and $w_{o,n} = p_n$. Set $f_{j,n} = \alpha_j(f_n)$.

The space K lies in an arbitrary universe; it need not be separable. The vector-functional equality is assumed for every $a \in A$. Cyclicity of ξ in all of K is not assumed, nor is irreducibility of σ . Inner products are linear in their second argument. Here $B(K)$ is the algebra of bounded complex-linear operators on K .

Conclusion

Each fixed vector $\sigma(b)\xi$ is sent to zero in the limit. The estimate $\|\sigma(f_{j,n})\sigma(b)\xi\|^2 \leq \|b\|^2 2^{-n}$ proves this directly.

This declaration is a statement about orbit vectors and norm convergence of their images. It does not claim operator-norm convergence of the projections, or convergence on every vector of an arbitrary ambient representation without an additional argument.

Proof route

For a projection p , the vector-functional hypothesis gives $\|\sigma(p)\sigma(b)\xi\|^2 = \tau(b^*pb)$. Traciality rewrites this as $\tau(pbb^*p)$. Since $0 \leq pbb^*p \leq \|b\|^2 p$, positivity gives the bound $\|b\|^2 \tau(p)$. Apply it to $p = f_{j,n}$, whose trace is 2^{-n} by the transported-flag trace theorem, and let n tend to infinity.

▼ Proof steps

1. Use $p^* = p = p^2$ and star preservation to express the squared norm as $\tau(b^*pb)$.
2. Use the trace identity and the positive order bound on pbb^*p to obtain $\|\sigma(p)\sigma(b)\xi\|^2 \leq \|b\|^2 \tau(p)$.
3. Insert the exact value $\tau(f_{j,n}) = 2^{-n}$. The nonnegative squared norms are squeezed to zero, which implies norm convergence of the vectors.

Main citations

- Exact statement and proof
- A fixed family of compatible shell data
- The root flag transported by the chosen automorphism
- Transported flags are self-adjoint projections
- Exact trace values of transported flags
- Positivity of the completed trace
- The trace identity
- Projection-orbit estimate from a trace vector
- Trace decay implies orbit-vector convergence
- The continuous trace on the completion

Lean source signature (exact)

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theorem tendsto_transportedFlag_orbit_zero (family : RepresentativeShellFamily)
  (i : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit)
  (σ : Representation Limit H) (ξ : H)
  (hξ : ∀ a, Representation.vectorFunctional σ ξ a = trace a) (b : Limit) :
  Tendsto (fun n ↦ σ (transportedFlag family i n) (σ b ξ)) atTop (nhds 0)
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In the source	Mathematical meaning
family; i : GNSClass Limit	The given shell family and fixed class j (i in the source), supplying $f_{j,n} = \alpha_j(f_n) \in A$ at every natural stage.

In the source	Mathematical meaning
$\sigma : \text{Representation Limit } H; \xi : H$	The unital star representation $\sigma : A \rightarrow B(K)$ on the complete complex Hilbert space denoted H in the source, and a vector $\xi \in K$. Neither irreducibility nor cyclicity in all of K is assumed.
$h\xi : \forall a, \text{Representation.vectorFunctional } \sigma \xi a = \text{trace } a$	For every $a \in A$, $\langle \xi, \sigma(a)\xi \rangle = \tau(a)$, with inner product linear in the second entry. This is the trace-vector hypothesis on the given ξ .
$b : \text{Limit}; \sigma b \xi$	A fixed element $b \in A$ and the orbit vector $\sigma(b)\xi \in K$.
$\text{fun } n \mapsto \sigma (\text{transportedFlag family i n}) (\sigma b \xi)$	The vector sequence $n \mapsto \sigma(f_{j,n})(\sigma(b)\xi)$: the projection acts on the fixed orbit vector, not on b in the algebra.
$\text{Tendsto } (...) \text{ atTop } (nhds 0)$	As the natural number $n \rightarrow \infty$, this sequence tends to the zero vector in the norm topology of K . This conclusion is for each fixed b ; it does not assert operator-norm convergence or convergence on every vector of an arbitrary ambient K .

Further source notes: The filter expression asserts that $\sigma(f_{j,n})\sigma(b)\xi$ tends in norm to zero as $n \rightarrow \infty$, for each fixed b ; it makes no additional cyclicity assertion.

Earlier published Card and PDF

▼ Exact content identity

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